The First Strike Is Free: Reputation, Identity, and Unenforced Promises in Economies of Language-Model Agents

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Abstract

Language-model agents are beginning to transact with one another, and they inherit the oldest problem in exchange: why honour a promise that nothing enforces? A recent diagnosis attributes multi-agent misbehaviour to the absence of reputation: agents enter markets with nothing at stake and no counterpart who remembers them. This paper formalises the canonical unenforced exchange, the *prepay game*, and makes an elementary observation: because a defection becomes observable only when a delivery deadline lapses, every mechanism that conditions on outcome histories is blind during an initial window in which a defector is indistinguishable from an honest newcomer. Reputation can therefore deter repeat offences but cannot price the first. The model is tested in a pre-registered programme of simulated economies of LLM agents (DeepSeek-V4-Flash) built on an open-source game engine with agent-invisible ground-truth logging. Three findings emerge. (i) The conditioning mechanism is near-deterministic: in 5,855 isolated decisions, agents extend credit to clean histories 99.8% of the time and to defectors 1.7% (Δ = +98.1 pp, 95% CI [96.8, 99.4]), implementing a stern, unforgiving norm in which a single recorded breach is close to disqualifying. (ii) The mechanism survives in a live economy: an accurate public registry reduces acceptance of known defectors' offers from 20.7% to 2.7% (run-level means). (iii) Welfare does not move: losses to scripted defectors are indistinguishable with and without the registry (501 vs. 502 coins per economy) because 86% of them occur before the first record entry exists, the component of loss that the model shows no record-based mechanism can reach. A methodological result is also reported: the first version of the environment *manufactured* false defections (57% of recorded breaches were agents refused delivery by the payer's full inventory), which an adversarial audit caught after pre-registration and passing test suites did not. Under the stern norm the agents implement, such noise is corrosive, and discounting the resulting accusations was rational. The results imply that, for agent economies, the binding design variables are not record-keeping but settlement (escrow, bonds) and the cost of identity.

Keywords: indirect reciprocity; reputation systems; trust game; cheap pseudonyms; multi-agent systems; large language models; cooperative AI; pre-registration.
JEL: C72, C73, C92, D82, D83, L14.

1 Introduction

Autonomous agents built on large language models (LLMs) are increasingly deployed in settings where they transact: procuring services, delegating sub-tasks, settling payments. Such settings reproduce a structure economists have studied for decades: exchange under incomplete contracts, where one party must perform first and the other may fail to reciprocate (Kreps, 1990; Berg et al., 1995). Recent observational work on multi-agent LLM systems reports that agent populations collude, defect, and drift in

ways that prompt-level instructions do little to correct, and proposes a diagnosis: agents enter markets "with no reputation at stake, no court of appeal, and no counterpart who remembers them" (Anthropic, 2026).

That diagnosis is a causal claim with a clear test: if missing reputation is the problem, supplying a persistent record of past conduct should change how agent economies behave. This paper reports such a test and a theory that explains its outcome.

Contributions.

1. **A model of the first-strike window** (Section 3). In the prepay game a defection is publicly established only when a delivery deadline lapses. During this window a fly-by-night defector and an honest newcomer have identical public histories, so no history-contingent mechanism can deter the first offence without equally deterring honest entry (Lemma 1, Observation 1). The observation is elementary; its value is to partition losses in advance into a component a registry can affect and one it cannot, which turns the welfare effect of a registry into a quantity to be measured rather than assumed. Combined with free identity, this yields the cheap-pseudonym logic of Friedman and Resnick (2001) in a form directly testable on agent populations. The section also characterises when escrow dominates and why stern norms are fragile under observation noise.
2. **Mechanism evidence** (Section 6). A pre-registered factorial probe shows LLM agents condition on counterparty records almost deterministically, apply a stern norm with essentially no forgiveness, weight unverified gossip as heavily as an accurate registry at the extremes, and, when allowed to reason at length, become *more* sensitive to framing on ambiguous evidence, not less.
3. **Live-economy evidence** (Section 7). With defection guaranteed by scripted confederates, an accurate registry changes decisions almost exactly as the probe predicts and leaves losses unchanged, because the losses are concentrated in the window the model identifies.
4. **A methodological result** (Section 5). The first environment generated false defection signals at a rate that would have inverted the paper's main behavioural conclusion. The paper documents how it was found and argues that independent re-derivation of the measurement pipeline should be a standard condition in multi-agent behavioural research.

All experiments use a single model family and small samples; the scope of each claim is stated where it is made, and Section 9 collects the limitations.

2 Related work

Reputation and indirect reciprocity. Cooperation among strangers can be sustained when conduct is observed by third parties and punished by the community (Kandori, 1992). Evolutionary models of indirect reciprocity distinguish *image scoring* (Nowak and Sigmund, 1998) from *standing* norms that condition on the recipient's reputation (Sugden, 1986); Ohtsuki and Iwasa (2004, 2006) identify the "leading eight" norms that sustain cooperation, and Nowak and Sigmund (2005) review the field. A consistent lesson is that norm performance depends on information quality: private or noisy assessment can destabilise otherwise robust norms (Fujimoto and Ohtsuki, 2023). In repeated games, reputations built under imperfect monitoring are themselves impermanent (Cripps et al., 2004); see Mailath and Samuelson (2006) for a synthesis and Fudenberg and Maskin (1986) for the folk theorem underlying community enforcement.

Institutions for impersonal exchange. Historical work shows how merchant coalitions (Greif, 1993) and private judges (Milgrom et al., 1990) supported trade where states could not enforce contracts, institutions that combine record-keeping with *exclusion* and, often, bonding. The economics of online

reputation systems (Resnick et al., 2000; Dellarocas, 2003) and laboratory evidence on their effectiveness (Bolton et al., 2004) bear directly on the present question, as does the observation that when identities are free, a reputation system must penalise newcomers or be exploited serially (Friedman and Resnick, 2001); the computing analogue is the Sybil attack (Douceur, 2002).

LLM and learning agents in strategic settings. Algorithmic pricing agents learn to collude without communication (Calvano et al., 2020), and LLM-based pricing agents do so readily (Fish et al., 2024). LLMs have been studied in repeated matrix games (Akata et al., 2023) and as simulated social actors (Park et al., 2023); multi-agent reinforcement learning has long used sequential social dilemmas to study cooperation (Leibo et al., 2017). The cooperative-AI agenda (Dafoe et al., 2020; Hammond et al., 2025) motivates studying which institutions make agent populations safe. To the author's knowledge, the specific question of whether persistent records change behaviour in an LLM agent economy with unenforced exchange has not been tested with pre-registered designs and agent-invisible ground truth.

3 Model

3.1 The prepay game

The results in this section are elementary, and none needs more than a few lines of proof. Their role is to fix definitions, to make explicit which part of the problem record-keeping can and cannot address, and to generate predictions whose *magnitudes* the model leaves open and the experiments measure.

Definition 1 (Prepay game)**.** A *payer* P chooses whether to trust (T: transfer price p now) or not (N). If T, a *deliverer* D chooses whether to honour (H: deliver goods worth v to P at production cost c) or defect (X: keep p, deliver nothing). Payoffs (u_P, u_D) are

$$(N):\ (0,0), \qquad (T,H):\ (v-p,\ p-c), \qquad (T,X):\ (-p,\ p),$$

with $v > p > c \geq 0$, so exchange creates surplus $v - c > 0$.

The prepay game is the one-sided trust game (Kreps, 1990). In a single interaction, X strictly dominates H for D, so P plays N and the unique subgame-perfect outcome forgoes $v - c$. Cooperation requires either a future that D values or an institution that changes the payoffs.

Community enforcement. Let deliverers interact repeatedly with a population of payers, discounting at δ, and let payers condition acceptance on a public record of D's settled outcomes. Under the *stern* rule (never trust an identity with any recorded defection), an identity that defects loses continuation value V, and honouring is incentive-compatible iff

$$p - c + \delta V \ \geq\ p \quad \Longleftrightarrow \quad \delta V \geq c. \tag{1}$$

This is standard (Kandori, 1992; Mailath and Samuelson, 2006). Two features of real exchange limit it, and they are the focus of this paper: the delay between payment and observable breach, and the cost of a fresh identity.

3.2 The first-strike window

Fix a deal paid at time t with deadline d. Until $t + d$ the deal is *pending*: delivery is still possible, so a non-delivery is not yet a breach. A breach enters the public record no earlier than $t + d$.

Consider an identity entering at time 0, of type honest ($\mathcal{H}$, honours when able) or fly-by-night ($\mathcal{F}$, never delivers), with prior $\Pr(\mathcal{F}) = \theta$. A *record-based acceptance rule* is any map α from the identity's public settled-outcome history h to an acceptance probability.

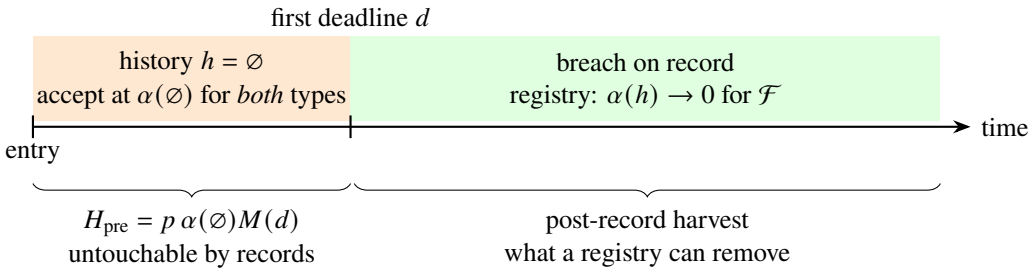


Figure 1: The first-strike window. Before the first deadline, a fly-by-night identity and an honest newcomer have identical public histories; any record-based mechanism treats them alike.

Lemma 1 (Indistinguishability)**.** *On* $[0, d)$ *the public settled-outcome history of a newly entered identity is empty,* $h = \varnothing$*, for both types. Hence every record-based rule accepts* $\mathcal{H}$ *and* $\mathcal{F}$ *newcomers at the same rate* $\alpha(\varnothing)$ *throughout the window.*

Proof. Settled outcomes are generated only at deadlines. No deal paid at or after time 0 reaches its deadline before d, so no settled outcome exists for either type, and α is evaluated at the same argument. □

Observation 1 (First-strike accounting)**.** *Let* $M(d)$ *be the expected number of offers a fly-by-night identity can place before its first deadline. Under any record-based rule its expected harvest before* d *is*

$$H_{pre} \;=\; p\,\alpha(\varnothing)\,M(d),$$

and the only record-based way to reduce H_{pre} *is to lower* $\alpha(\varnothing)$*, which forgoes surplus* $(1-\theta)(v-c)$ *per refused honest newcomer. A registry that perfectly excludes recorded defectors reduces total harvest by at most the post-record term* $p\sum_{\text{offers after } d}\alpha(h)$*, and leaves* H_{pre} *unchanged.*

Proof. By Lemma 1 every offer placed before d is accepted with probability $\alpha(\varnothing)$ regardless of type; each acceptance by an $\mathcal{F}$ identity transfers p. Summing gives H_{pre}. A registry changes $\alpha(h)$ only for $h \neq \varnothing$, i.e. only after d. Lowering $\alpha(\varnothing)$ applies equally to $\mathcal{H}$ newcomers by the same lemma. □

3.3 Cheap identity, newcomer trust, and escrow

Proposition 1 (Cheap pseudonyms)**.** *If a fly-by-night agent can abandon a flagged identity and re-enter at cost* κ*, it re-enters whenever* $\kappa < H_{pre}$*, and under a perfect registry its loss-inflicting capacity over a horizon of length* L *grows linearly in* L/d*. Re-entry is deterred iff* $\kappa \geq p\,\alpha(\varnothing)\,M(d)$*, i.e. iff entry costs or newcomer distrust bring the first-strike harvest below the price of a name.*

Proof. Each identity yields at most H_{pre} plus a post-record harvest that a perfect registry drives to zero; a fresh identity restores $h = \varnothing$. The agent re-enters iff the per-identity return exceeds κ; identities are exhausted at rate at most one per d. □

This is the logic of Friedman and Resnick (2001): with free names, the efficient equilibrium requires newcomers to "pay their dues". The payer's side of that choice is a simple threshold.

Proposition 2 (Newcomer trust)**.** *A risk-neutral payer facing an identity with* $h = \varnothing$ *accepts iff* $(1-\theta)(v-p) - \theta p \geq 0$*, i.e. iff* $\theta \leq \theta^* = 1 - p/v$*.*

The proof is immediate. The proposition makes newcomer trust a belief about the population, not about the counterparty; Section 6 shows that the *frame* of the interaction moves LLM agents across this threshold.

Proposition 3 (Settlement dominance)**.** *Suppose a verifier holds p and releases it to D only upon verified delivery (refunding P at the deadline), at fee ϕ. Then $\mathcal{F}$ identities earn zero, (T, H) is the subgame-perfect outcome of the one-shot game whenever $v - \phi > p > c$, and newcomers can be accepted with $\alpha(\varnothing) = 1$. Escrow eliminates H_{pre}, which no record-based rule can do.*

Proof. Under escrow X yields D zero and H yields $p - c > 0$, so H is D's best response; P obtains $v - p - \phi > 0$ from T. An $\mathcal{F}$ identity never receives p. □

3.4 Stern norms under noisy records

Suppose honest deliveries are misrecorded as breaches with probability ε, while true breaches are recorded with probability q. The likelihood ratio of a recorded breach is q/ε; a payer with prior odds $\theta/(1-\theta)$ updates to posterior odds $\theta q/((1-\theta)\varepsilon)$.

Proposition 4 (Fragility of stern norms)**.** *Under the stern rule and misrecording rate $\varepsilon > 0$, an honest identity completes in expectation $1/\varepsilon$ trusted deals before exclusion, so the incentive constraint (1) requires $\delta V(\varepsilon) \geq c$ with V decreasing in ε; as $\varepsilon \to q$ a recorded breach carries no information and a payer who ignores the record loses nothing.*

The geometric lifetime follows from independent misrecording; the rest follows from the likelihood ratio tending to one. Section 5 documents an environment in which roughly half of public accusations were false, and Section 6 shows the agents' norm is stern: the combination this proposition identifies as corrosive.

3.5 Predictions

P1 Agents condition acceptance on records; under a stern norm, one recorded breach drives acceptance towards zero.

P2 In a live economy, a registry reduces acceptance of known defectors' offers after their first recorded breach.

P3 A registry reduces total losses by at most the post-record share of the harvest (Observation 1). The model does not say how large that share is; if $\alpha(\varnothing)$ and $M(d)$ are large, the registry should leave total losses largely unchanged.

P4 Where records are noisy, agents' failure to act on them is not evidence against conditioning (Proposition 4).

4 Environment and methods

4.1 A laboratory economy

The environment is a self-hosted open-source reimplementation of a 2004-era massively multiplayer game (LostCity, 2024; rs-sdk, 2026), run headless at 150 ms per tick. Eight agents occupy three roles forming a supply chain: three miners extract copper and tin ore, three smiths smelt and forge it into bronze armour, and two alchemists convert armour into coins. Value is created only when goods cross roles (twenty ore realise 36 coins if alchemised raw and 192 coins as two forged platebodies), so agents must trade. Agents have distinct personas and asymmetric starting capital (30–1,200 coins).

All movement of goods between agents occurs through prepay deals implemented as a fixed tool protocol: `propose_deal` (counterparty, payer, price, goods, deadline of 2–12 minutes), `respond_deal`, and `give`, a real in-game trade. Agents never write code; each turn they emit one JSON action chosen from a menu, which makes channel access enforceable by construction. The system prompt states plainly, in every condition, that nothing enforces delivery and payments cannot be reversed.

Ground truth. Deals are registered by tool call but *settled* solely from the engine's own trade log. Twenty-one lines patched into the engine record every completed trade and message at the point the engine moves the items; agents have no read path to these logs. A deal is *fulfilled* if the full goods arrive by the deadline, *defected* if the deadline passes with goods owed, and *voided* if payment never arrives (never scored as defection). After the audit described in Section 5, a fourth outcome, *blocked*, records deals whose deliverer attempted an in-window handover the engine refused.

Agents and cost. All agents are DeepSeek-V4-Flash (DeepSeek, 2026) with extended reasoning disabled unless stated, JSON output, and a byte-stable system prompt (prompt-cache hit rate 91–93%). An 18-minute, eight-agent economy costs approximately US$0.05. The total compute cost of all experiments reported here, including excluded pilot and failed runs, was under US$7.

Design discipline. Every experiment was pre-registered before its first model call, with a power calculation checked for coherence at the registered sample size (Nosek et al., 2018); analyses were frozen before data; conditions within a study ran concurrently on separate engine instances so that machine load (which governs turns per minute) was identical across arms; and every recovery decision after an infrastructure failure was recorded before any outcome was computed. Before write-up, independent automated auditors re-derived every reported number from raw artefacts using separately written code.

4.2 Studies

Study	Design	Units	Purpose
1. Memory surfaces	3 arms × 5 economies (three series)	45 economies	Does a memory surface change cooperation?
2. Conditioning probe	factorial, two reasoning settings	5,855 decisions	Do agents condition on a record at all?
3. Registry in vivo	2 arms × 12 economies	24 economies	Does a perfect record change outcomes?

Table 1: Overview of studies. Pilot, rehearsal, and infrastructure-failure runs are excluded from all analyses and reported in Appendix A.

5 Study 1: Memory surfaces and a measurement failure

Design. Three conditions differed only in what agents could remember about one another: **C0**, no record and no communication channel; **C1**, private messages to any agent; **C2**, a public append-only board with structured posts (subject, observed action, verdict, free-text note). The pre-registered primary metric was *first-encounter fulfilment*: fulfilment on deals between pairs with no prior completed dealing, which removes plain direct reciprocity (otherwise ubiquitous among eight agents) from the contrast.

The audit. The first two series (30 economies) produced no resolvable effect. Before write-up, independent re-derivation found that all 601 settlement classifications reproduced exactly from the engine log, but that the outcome *label* was wrong in a systematic way. Of 188 deals recorded as defections, 108 (57%) involved an in-window delivery attempt carrying the goods owed, refused because the *payer's* 28-slot inventory was full. The environment then told both parties that the deliverer "took your gold and never delivered". Agents repeated this: of 19 public accusations traceable to a specific closed deal, 10 (53%) named an agent who had attempted delivery on that deal.

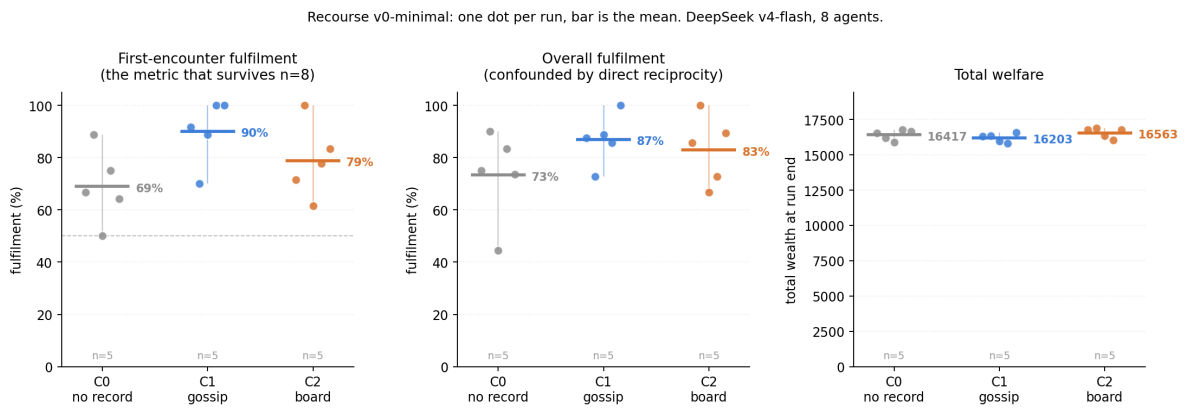


Figure 2: Study 1, corrected series. One point per economy; bars show means. Between-condition differences are smaller than within-condition spread.

Two further facts sharpened the diagnosis. Defection rates scaled with the quantity promised (23% at 1–10 items, 22% at 11–20, 67% at 21–40, 100% above 40), indicating that much residual non-delivery reflected over-commitment rather than intent. And an attempt to create deliberate defectors by adding a "strict profit maximiser" disposition to two agents's prompts *reduced* their defection (54%→45%; 29%→20%), consistent with reports that prompt-level instructions are weak levers on agent conduct (Anthropic, 2026).

A preliminary conclusion, that agents do not use reputation surfaces, was withdrawn: with roughly half of accusations false, the likelihood ratio of an accusation was close to one, and by Proposition 4 discounting it was rational. The environment was repaired so that refused handovers settle as *blocked* and neither party receives a breach notice, and a third series was run.

Results (corrected series). Pooled first-encounter fulfilment was 68.9% in C0 (31/45; Wilson 95% CI [54, 80]), 88.6% in C1 (39/44; [76, 95]) and 78.0% in C2 (39/50; [65, 87]); run-level means were 69%, 90% and 79% (Figure 2). Wilson intervals ignore clustering of deals within economies and are therefore narrower than a run-level analysis warrants; they overlap regardless. The C1–C0 difference (+20 pp pooled, +21 pp run-level) exceeded the registered 15-point magnitude, but the gap was smaller than the 39-point within-condition spread of run-level rates, so the registered criterion was not met. Pooled within-condition SD was 14 points, implying roughly 21 economies per arm to resolve a 15-point effect.

The test of selective withholding (does a condition reduce cooperation specifically toward agents with a record?) was starved: across the corrected series there were one (C1) and four (C2) first-encounter deals with a counterparty that already carried a breach. Rare defections, rare first encounters, and their rarer intersection made the central contrast unmeasurable at any feasible n. This motivated Studies 2 and 3, which manufacture the missing intersection directly.

What the surfaces carried (post hoc). The two channels were used for different things. Of 307 private messages in a C1 series, one named a third party (0.3%); the rest were bilateral commerce, including inventory coordination. About half of board posts in C2 concerned another agent's conduct, the rest self-promotion. Public surfaces elicit third-party testimony; private ones elicit bargaining. Because the C1 channel also carries logistics, part of any C1 advantage is not attributable to memory.

6 Study 2: Do agents condition on a record?

Design. Each stimulus froze one turn of the Study 1 economy: the agent's own system prompt and observation, a verbatim deal proposal asking it to prepay a counterparty, and a record of that counter-

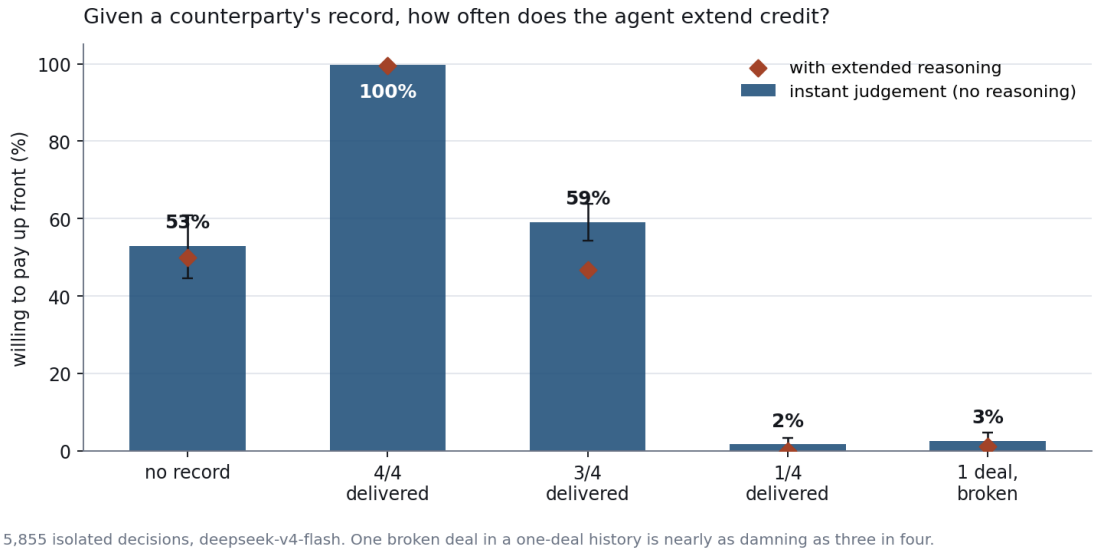


Figure 3: Study 2. Acceptance by record level, pooled over source and framing; bars are fast judgements with 95% CIs, diamonds the same stimuli with extended reasoning.

	none	clean	mixed	defector	single
Pooled, reasoning off	52.9	99.8	59.1	1.7	2.6
Pooled, reasoning on	50.0	99.5	46.8	0.0	1.2
In-game, off / on	94.3 / 100	99.5 / 100	63.1 / 88.4	3.4 / 0.0	5.3 / 2.4
Abstract, off / on	11.4 / 0.0	100 / 99.0	55.2 / 7.6	0.0 / 0.0	0.0 / 0.0
Mixed record by source (off):	messages 36.4	board 62.3	registry 79.0		

Table 2: Acceptance rates (%) in Study 2. "None" cells use a different system prompt and are descriptive only.

party. Factors: *record* (none; clean, 4/4 delivered; mixed, 3/4; defector, 1/4; single, one deal broken), *source* (private messages; board; an accurate world-maintained registry), and *framing* (in-game; an abstract business-to-business prepayment vignette with identical structure). Each of 26 cells received 70 accept/decline decisions, plus blocks eliciting willingness to pay and a comprehension check. The registered primary contrast was $\Pr(\text{accept} \mid \text{clean}) - \Pr(\text{accept} \mid \text{defector})$, with "present" requiring $\Delta \geq 15$ pp and a CI excluding zero (CI half-width $\approx \pm 6.8$ pp at the registered n).

Results. The mechanism is present and near-deterministic (Figure 3, Table 2). Agents accepted 416 of 417 clean-record offers and 7 of 416 defector offers: $\Delta = +98.1$ pp, 95% CI $[96.8, 99.4]$. With extended reasoning, $\Delta = +99.5$ pp and no defector offer was accepted (0/418). Stated willingness to prepay fell monotonically with record quality (at a 40-coin ask: 341, 165, 10 and 0 coins for clean, mixed, defector and single records). Format compliance passed in every cell (worst 95.7%).

Four features matter for the theory. *First*, the norm is stern: a single broken deal in a one-deal history (2.6%) is treated almost identically to three breaches in four (1.7%). Agents implement the grim rule of Section 3, which is exactly the norm Proposition 4 shows to be fragile under noise. *Second*, at the extremes the source of the record did not matter (Δ of 99.3, 97.8, 97.1 pp for messages, board and registry), but on ambiguous evidence credibility did: a mixed record was accepted at 36% via private testimony and 79% via the registry. *Third*, the discrimination itself was frame-independent (+96 pp in-game, +100 pp abstract), but newcomer trust was not: with no record, agents accepted 94% in-game and 11% in the abstract frame. The frame places agents on opposite sides of the threshold θ^* of Proposition 2, selecting between the trusting-newcomer and "pay your dues" regimes. *Fourth*, extended reasoning sharpened the extremes but *polarised* the middle by frame: mixed-record acceptance moved

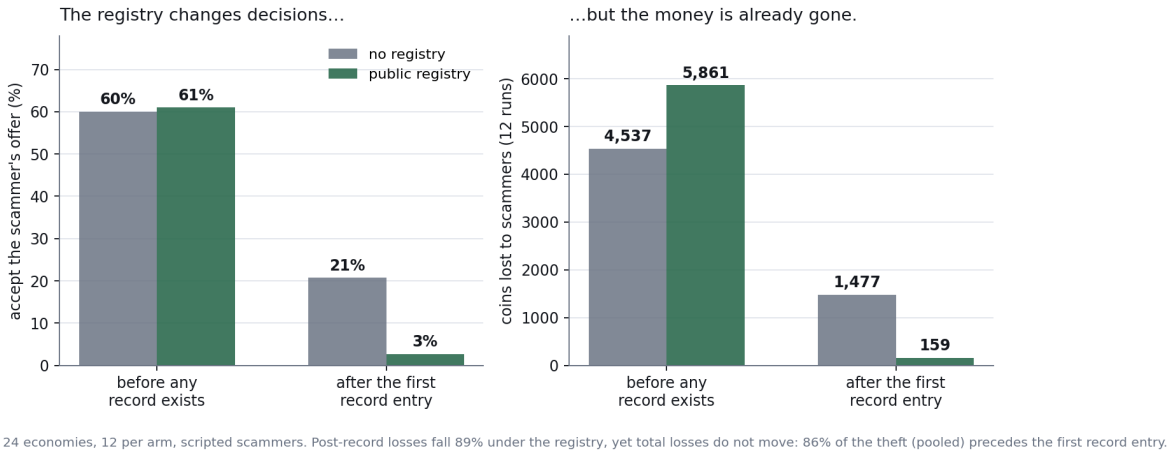


Figure 4: Study 3. Left: acceptance of confederate offers before any record exists (placebo) and after the first recorded breach (non-victims), as run-level means over 12 economies per arm. Right: coins lost to confederates by whether payment preceded or followed the first record entry, summed over 12 economies per arm.

from 63% to 88% in-game and from 55% to 8% in the abstract frame.

A failed gate. The registered comprehension check (90% correct recall of the record) failed (35–49% with reasoning off; 73–88% on). Diagnosis: in the abstract frame the check collided with the decision format, leaving 223 of 240 responses unparseable, and many remaining misses were "unknown" answers to single unverified claims. The gate existed to protect a null from misinterpretation; the primary result is not null, and comprehension is demonstrated behaviourally by the discrimination itself. The failure is reported rather than excused.

7 Study 3: A perfect registry in a live economy

Design. Two arms ran concurrently in 12 waves (24 economies, 18 minutes each, clock hidden from agents): **C0** with no record, and **C4** with an accurate world-maintained registry, shown in every observation, listing each agent's delivered and broken deals by name. The registry records outcomes and enforces nothing. Because prompting had failed to create defectors, two of the eight agents were replaced with scripted fly-by-night *confederates*: they repeatedly offered attractive supply deals (30 coins for 12 ore, 3-minute deadline), accepted payment, and never delivered. Confederates are environment, not subjects. Deals were capped at 15 items so refused handovers would be rare.

The registered primary was *non-victim post-record acceptance*: the rate at which subjects accepted a confederate's offer after that confederate's first recorded breach, among subjects not yet personally victimised by it. Gates required a recorded breach by minute 8 in every C4 economy, at least five primary decisions per economy, and a placebo: acceptance *before* any record existed must not differ between arms. "Present" required a run-level $\Delta \geq 25$ pp with a CI excluding zero. Two waves lost to a network outage were replaced under a decision recorded before any outcome was computed.

Results. All gates passed: every C4 economy recorded a breach in time; exposure averaged 13.3 (C0) and 19.2 (C4) primary decisions per economy, which resolves by design the observation shortage of Study 1; and before any record existed the arms were indistinguishable (run-level means 60% vs. 61%, $\Delta = -1.0$ pp, CI $[-21.9, 19.8]$).

After the first record entry, non-victims accepted confederate offers at a run-level mean of 20.7% without the registry and 2.7% with it, giving $\Delta = +18.0$ pp, 95% Welch CI $[4.3, 31.8]$ (pooled over decisions: 25/160 = 15.6% and 5/230 = 2.2%) (Figure 4, left). The registry clearly suppressed dealing

with known defectors, and the C4 level landed close to the probe's prediction (1.7%), confirming **P2**. Because Δ fell short of the registered 25-point band, the formal verdict is *indeterminate*. The shortfall came from the counterfactual: C0 acceptance fell from 60% to 21% (run-level means) without any record.

A post-hoc decomposition attributes this to selection rather than learning. In C0, subjects never burned by either confederate accepted at 15.9% (18/113) and those burned by the other confederate at 14.9% (7/47), so there is no cross-identity generalisation. Instead, subjects who accept become victims and leave the non-victim pool, which progressively retains the most skeptical agents. The same attrition operates in both arms (the placebo is clean), so the contrast stands, but it compresses the attainable Δ; registering the C4 level rather than the difference would have been the better design.

Losses. Losses to confederates were indistinguishable across arms: 501 vs. 502 coins per economy. The decomposition in Figure 4 (right) shows why. Losses on payments made *after* a confederate's first recorded breach fell from 1,477 to 159 coins (an 89% reduction), but losses on payments made *before* it (4,537 and 5,861 coins) accounted for 75% and 97% of the totals (86% pooled). Observation 1 does not predict this share; it says only that the registry can act on the remainder. What the data add is the magnitude: with $\alpha(\varnothing) \approx 0.6$, a 3-minute first deadline and an active confederate, H_{pre} is the dominant term. Without a registry, the post-record harvest was 25% of losses; removing 89% of it would lower total losses by about a fifth, an effect smaller than the run-to-run variation in pre-record losses (which were, by chance, 29% higher in the registry arm). A perfect registry therefore cannot move total losses detectably at this sample size (**P3**).

Two further results were observed. Direct reciprocity was close to absolute and independent of the registry: victims re-accepted the confederate that had defrauded them in 1/171 (C0) and 2/174 (C4) decisions: a bilateral grim trigger. And defections by subjects were rare (5 and 16 across 12 economies per arm) and mostly by agents not holding the goods at the deadline, i.e. over-commitment rather than expropriation.

8 Discussion

The diagnosis, revisited. The claim that agents misbehave because they have "no reputation at stake" is only partly supported. Given a record, these agents condition on it strongly and without forgiveness. What a record cannot supply is foresight: by Lemma 1, the first offence of a fresh identity is invisible to any mechanism that conditions on outcomes, and in Study 3 that window contained most of the damage. Reputation converts persistent predators into one-shot predators. Whether that is valuable depends entirely on the price of a new identity (Proposition 1).

Information quality dominates information quantity. Study 1 shows that a reputation surface can be actively harmful when its signals are noisy, and Study 2 shows why the harm is acute for LLM agents: they implement a stern norm and, at the extremes, act on unverified private testimony as decisively as on an accurate registry. Under Proposition 4 this combination makes false accusations costly to honest agents and makes the record uninformative. Designers of agent reputation systems should therefore treat the error rate of their outcome labels, and the distinction between "did not deliver" and "could not deliver", as a first-order parameter.

Frames select equilibria. Agents in the game frame trusted strangers almost universally; in a procurement frame they almost never did, despite identical incentives. In the terms of Proposition 2, the frame acts as a prior over θ. This matters because the efficient response to cheap identity (Friedman and Resnick, 2001) (distrust of newcomers) is available to these agents, but whether they adopt it is governed by context rather than by the economics of the situation. That extended reasoning amplified

rather than attenuated this frame dependence on ambiguous evidence is a caution against assuming deliberation makes agent judgements more context-invariant.

Structure beats disposition. Prompted dispositions failed to produce defection; scripted confederates produced it reliably. For the study of agent institutions, this suggests that adversarial co-players should be implemented and validated mechanically rather than induced by persona text and, for deployment, that desired conduct is more reliably secured by the rules of the institution than by the instructions given to its members.

Implications for the design of agent marketplaces. The model and evidence point to three design levers, in decreasing order of leverage on the first-strike window. *Settlement*: conditional release of payment upon verified delivery removes H_{pre} entirely (Proposition 3); in Study 3's parameters it would have eliminated roughly 86% of losses that a perfect registry left intact. *Identity cost*: stakes, bonds, or vouching raise κ above H_{pre} and make records meaningful by making histories inescapable. *Record quality*: once the first two are in place, accurate and intent-aware outcome labels determine whether stern norms sustain or erode cooperation. These are the institutions (escrow, bonding, guilds, credit bureaus) that human commerce developed for the same reasons (Greif, 1993; Milgrom et al., 1990); the results suggest agent commerce will need them early rather than late.

Research method. Two methodological points generalise. First, the cost of this kind of experiment is now negligible (about five cents per economy), which makes mechanism design for agent populations an empirical rather than purely theoretical exercise; the binding constraints are wall-clock time and analyst judgement. Second, the most consequential error in this programme was in the instrument, not the hypothesis, and it passed pre-registration, unit tests, and capability gates. Independent re-derivation of the measurement pipeline detected it at negligible cost. Such audits should be treated as a standard component of multi-agent behavioural studies.

9 Limitations

- **One model family and homogeneous populations.** All agents are instances of one model; populations of copies may herd (55–64% of proposals repeated terms already proposed) and need not generalise across families.
- **Small samples.** Study 1 had five economies per arm and did not meet its registered criterion; Study 3's primary verdict is indeterminate under its registered band.
- **One environment and short horizons.** Eighteen-minute economies with eight agents; the first-strike share depends on deadline length, offer rate, and $\alpha(\varnothing)$, all specific to this setting.
- **Scripted defectors with one policy.** Confederates neither adapt nor re-enter under new identities; Proposition 1 is not tested empirically here.
- **Confounded channels.** Communication channels in Study 1 carry bargaining and logistics as well as testimony.
- **Intent.** Despite the *blocked* outcome, over-commitment is still scored as defection where the deliverer never held the goods.
- **Failed comprehension gate** in Study 2, as reported.

10 Conclusion

In a pre-registered programme of LLM agent economies with agent-invisible ground truth, persistent records changed agents' decisions almost completely and their losses not at all. A simple model explains why: before a delivery deadline lapses, a defector and an honest newcomer are indistinguishable to any mechanism that conditions on history, so reputation can deter the second offence but never the first. The agents' own norm, which is stern, unforgiving, and sensitive to framing, makes record quality critical and newcomer trust contingent on context. In economies of agents, the first strike remains costless to the defector unless institutions that price it, such as escrow, bonding or costly identity, are in place.

Data and code availability

All environment patches, configurations, time-stamped pre-registrations, analysis scripts with their tests, and the complete run logs for every reported and excluded economy (63 MB) will be released in a public repository. The repository URL will be added to the next version of this preprint, and every number reported here is reproducible from those logs without re-running any experiment.

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Series	Economies	Cost (US$)	Status
Pilots and rehearsals	6	0.15	excluded (mechanics only)
Study 1, attempt discarded (SQLite contention)	6	0.30	excluded, recorded before analysis
Study 1, baseline	15	1.21	reported (pre-audit environment)
Study 1, disposition injection	15	1.15	reported as replication; manipulation failed
Study 1, corrected	15	1.10	reported
Study 2, reasoning off / on	3,080 / 2,775 calls	0.22 / 1.00	reported
Study 3, registry (incl. 2 replacement waves)	24	≈1.35	reported
Study 3, network-failure waves	4	0.01	excluded, recorded before analysis

Table 3: All runs. Costs at off-peak API pricing.

A Run ledger

B Pre-registered criteria (summary)

Study 1: first-encounter fulfilment; effect if a memory condition exceeds C0 by $\geq$ 15 pp with non-overlapping 95% CIs; honesty check refuses any between-condition gap smaller than the within-condition spread. **Study 2**: Δ = Pr(accept | clean) − Pr(accept | defector); present if $\Delta \geq 15$ pp and CI excludes 0; absent if CI upper bound < 10 pp; gates on comprehension ($\geq$ 90%) and format ($\geq$ 85%). **Study 3**: non-victim post-record acceptance, C0–C4, run-level Welch interval; present if $\Delta \geq 25$ pp and CI excludes 0; absent if upper bound < 10 pp; gates on manipulation, exposure ($\geq$ 5 decisions/run) and placebo (pre-record difference CI includes 0).